



Improving Metadata and Discoverability

Making your software findable

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Findability comes from description

A Zenodo record with a bare title and no keywords is a catalog record with no subject headings: retrievable if you already know it exists, invisible if you don't.

The systems are GitHub topics, Zenodo fields, and a README instead of a catalog. The judgment — *what would a seeker search for, and does this record contain it* — is unchanged.

In earlier episodes you created a CITATION.cff, a license, a repo structure, and a Zenodo DOI. **This episode ties them together.**

What counts as useful metadata?

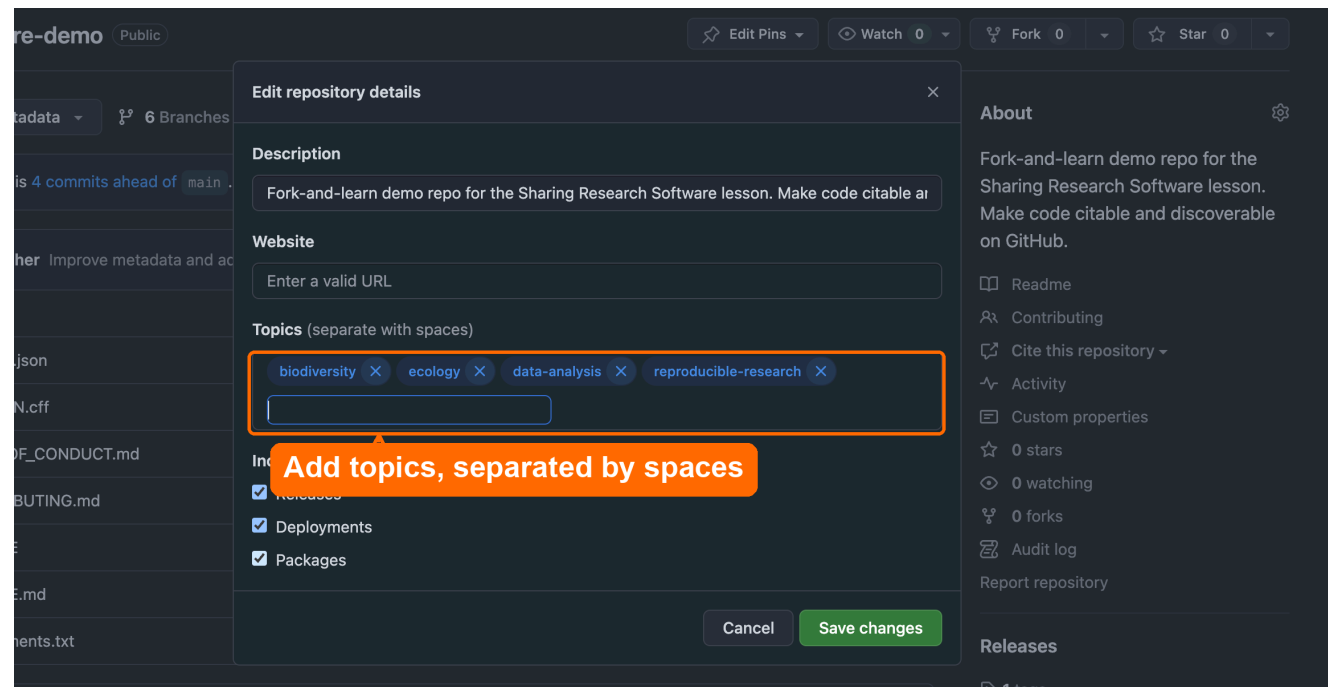
Good metadata answers predictable questions with minimal effort from the reader:

- **What is this software?** — short description or abstract
- **Who made it?** — authors, ORCIDs
- **How do I cite it?** — CITATION.cff + DOI
- **What domain is it for?** — keywords
- **What else does it relate to?** — related article, dataset, funding source

GitHub Topics improve search ranking

Add these in `Settings → General → Topics` :

- **Discipline tags** — `geospatial`, `text-mining`, `materials-science`
- **Method tags** — `simulation`, `visualization`, `machine-learning`
- **Language tags** — `python`, `r`



The 30-second rule

Your README is your software's front door.

If users can't understand what it does, how to install it, or how to use it in **30 seconds**, they leave.

A collection without a finding aid can only be used by someone with time to read every box. A repo without a real README can only be used by its author.

The 30-second rule is scope-and-content-note discipline applied to code.

README: 7 essential sections

1. **About** — what does this do? (2-3 sentences)
2. **Features** — key capabilities
3. **Getting Started** — prerequisites + install
4. **Usage** — minimal working example
5. **Citation** — link to CITATION.cff / DOI
6. **License** — state terms explicitly
7. **Contact** — how to get help

[UC OSPO README Guide](#) (*UC-specific*)

Before → after

✗ Before (`main` , starting state)

```
1 # Biodiversity Analysis Toolkit
2
3 Analysis tools for biodiversity
4 research data.
```

One line of description. No install steps, no usage, no citation, no license, no one to contact.

✓ After (`05-add-metadata` , abridged)

```
1 # Biodiversity Analysis Toolkit
2 [DOI badge] [License badge]
3 Analysis tools for biodiversity... <!-- 1 About -->
4
5 ## Features <!-- 2 -->
6 - Citable, open source, documented
7
8 ## Getting Started <!-- 3 -->
9 Prerequisites: Python 3.9+
10 ### Installation & Usage <!-- 4 -->
11 pip install -r requirements.txt
12 python src/analysis.py
13
14 ## Citation <!-- 5 -->
15 Cite via CITATION.cff or the DOI badge.
16 ## License <!-- 6 -->
17 BSD 3-Clause. See LICENSE.
18 ## Contact <!-- 7 -->
19 Open an issue in this repository.
```

Beyond the README: community health files

CONTRIBUTING.md How to contribute. Worth adding *if* you want contributions — or say plainly you don't.

CODE_OF_CONDUCT.md Behavioral standards. [Contributor Covenant](#) is the standard choice.

CHANGELOG.md Version history: what's new, what broke, what changed.

GitHub recognizes certain filenames

File	What GitHub does
README.md	Renders as the repository's landing page
LICENSE	Offers a license-template picker
CITATION.cff	Adds "Cite this repository" with APA/BibTeX
CODE_OF_CONDUCT.md	Counts toward the community profile
CONTRIBUTING.md	Links guidelines when someone opens an issue/PR
SECURITY.md	Becomes the security policy

Only appears after the file is on the default branch. Teach the *pattern* — the filename is a signal — not the exact button, since GitHub changes these over time.

Where should software live?

Zenodo is the recommended deposit location for software:

- Integrates directly with GitHub — archives each release automatically
- Mints a DOI for every version
- Flows into DataCite, Google Scholar, library catalogs
- Free, CERN-operated, widely recognized by journals and funders

Dataverse and Dryad are good for **datasets**, but don't auto-integrate with GitHub releases the same way.

UC-specific: all UC campuses have Dryad. If you deposit data there, Dryad prompts a linked Zenodo code deposit automatically.

Zenodo metadata is what discovery systems see

When you deposit on Zenodo, the record flows into **DataCite**, **Google Scholar**, and **library catalogs** (EBSCO, Primo, WorldCat).

Add or refine: authors + ORCIDs, keywords, related works, funding, a readable description.

Caution: if a repo has a `.zenodo.json`, Zenodo uses *that* file, not `CITATION.cff`. Keep them in sync or you'll publish two different citation records for the same software.

Supporting others: priority order when metadata is thin

1. A readable **description** — the biggest single loss when it's empty
2. **Author ORCIDs** — every J. Chen on Zenodo is indistinguishable without one
3. **Keywords**, reused consistently across GitHub, CITATION.cff, Zenodo
4. **Related works** — link the software to its paper, dataset, grant

Challenge: identify useful metadata



2 min

List three elements you'd add to improve a repository's discoverability.

Examples: ORCID IDs for each author · keywords describing domain and function · a link to a related article or dataset · funding acknowledgment

Challenge: improve your Zenodo record

 5 min

Open your Zenodo record from the earlier release episode. Select **Edit**, and add:

- keywords
- authors and ORCIDs
- a description
- related publication DOIs and funding

Save and publish the updated record when done.

Challenge: which change makes it discoverable?



3 min

A researcher's tool has been public on GitHub for a year. One-line README, no topics, Zenodo deposit with only a title. **Which single change matters most?**

A. Add collaborators B. Descriptive topics + matching keywords + a description C. More frequent commits D. A shorter name

B. Discovery happens through *description*, not access level or activity. This is the same reason a catalog record with no subject headings is nearly invisible.

Key points

- Metadata increases discoverability in GitHub, Zenodo, and scholarly indexes
- Keep information consistent across CITATION.cff, README, GitHub topics, and Zenodo — that's authority control
- Thoughtful metadata supports FAIR and helps others reuse your software